Human Nutrition & Metabolism 42 (2025) 200341

Contents lists available at ScienceDirect

Human Nutrition & Metabolism

journal homepage: www.sciencedirect.com/journal/human-nutrition-and-metabolism

Type 2 diabetes risk prediction using glycemic control Metrics: A machine learning approach

Radwan Qasrawi [a,b,*], Suliman Thwib [a], Ghada Issa [a], Razan Abu Ghoush [a], Malak Amro [a]

[a] Department of Computer Science, Al-Quds University, Jerusalem, Palestine
[b] Department of Computer Engineering, Istinye University, Istanbul, Turkey

ARTICLE INFO

Keywords:
Diabetes
Machine learning
Dietary metrics
Glycemic load
Glycemic index

ABSTRACT

Background: Type 2 Diabetes Mellitus (T2DM) remains a significant global health burden, particularly in low- and middle-income settings. Conventional prevention strategies often lack personalization, overlooking individual variability in lifestyle, nutrition, and health status. This study aimed to develop a personalized T2DM risk prediction model using machine learning (ML), integrating clinical, behavioral, and dietary data, including glycemic index (GI) and glycemic load (GL) derived from actual food and recipe intake.

Methods: Data from 3145 Palestinian adults (aged 18–60) were analyzed using statistical and machine learning (ML) techniques. Variables included age, sex, education, income, physical activity, smoking status, perceived health, and detailed nutritional intake, specifically glycemic index (GI) and glycemic load (GL). Nine ML models were developed using the AutoGluon-Tabular framework. Model performance was assessed via accuracy, area under the curve (AUC), and log loss. Feature importance analysis identified key predictors of T2DM risk.

Results: Women had significantly higher odds of diabetes than men, while rural residents had a lower risk compared to urban dwellers. People aged 50–59 were over six times more likely to be diabetic than those aged 18–29. Lower education and poor perceived health were also strong predictors. Diabetic participants consumed significantly lower GI (87.7 ± 36.1) and GL (241 ± 180.5) diets compared to non-diabetics (GI = 98.8 ± 35.5; GL = 303.3 ± 202.7; p = 0.001). Among the ML models, XGBoost and CatBoost performed best, with over 93 % accuracy and excellent prediction scores. Glycemic load, age, BMI, waist-to-hip ratio, and self-reported health status were the most important risk indicators.

Conclusion: This study showed the effectiveness of integrating machine learning with glycemic control metrics and lifestyle data for personalized T2DM prediction. Incorporating glycemic values from real food and recipe intake improved model accuracy and interpretability. These findings support the development of precision prevention strategies tailored to individual risk profiles, particularly in underserved populations.

1. Introduction

Type 2 Diabetes Mellitus (T2DM) is a long-term metabolic condition that results in chronically elevated blood glucose levels due to insulin resistance, impaired insulin secretion, or both. As one of the most serious global health concerns, T2DM currently affects more than 422 million people worldwide, a number that is expected to increase significantly in the coming decades [1]. The impact of this condition extends far beyond blood sugar levels; without proper management, T2DM can lead to serious complications such as cardiovascular disease, kidney failure, and vision loss [2].

Traditionally, glycemic control has been monitored using well-established biomarkers, most notably glycated hemoglobin (HbA1c), which offers a reliable snapshot of average blood glucose levels over a two-to three-month period [3,4]. However, recent attention has shifted toward dietary indicators, particularly the Glycemic Index (GI) and Glycemic Load (GL), as additional tools to support blood sugar regulation. The GI ranks foods based on how quickly they raise blood glucose after consumption, while the GL provides a more complete picture by accounting for both the quality and quantity of carbohydrates in a meal [5]. Research consistently shows that diets low in GI can help reduce post-meal glucose spikes and improve insulin sensitivity, making them a valuable component in dietary planning for individuals with diabetes [5,6].

Yet, managing T2DM goes beyond diet alone. It involves

* Corresponding author. Department of Computer Science, Al-Quds University, Jerusalem, Palestine.
E-mail address: radwan@staff.alquds.edu (R. Qasrawi).

https://doi.org/10.1016/j.hnm.2025.200341
Received 2 June 2025; Received in revised form 10 August 2025; Accepted 12 August 2025
Available online 14 August 2025

Abbreviation

T2DM	Type 2 Diabetes Mellitus
GI	Glycemic Index
GL	Glycemic Load
HbA1c	Glycated hemoglobin
BMI	Body Mass Index
ML	Machine Learning
KNN	K-Nearest Neighbors
SMOTE	Synthetic Minority Over-sampling Technique
AUC	Area Under the Curve
SFA	Saturated Fatty Acids
MUFA	Monounsaturated Fatty Acids
PUFA	Polyunsaturated Fatty Acids

understanding the interrelated association between factors that contribute to the disease, including genetics, behavior, lifestyle, and environment. This complexity makes T2DM difficult to predict and manage using conventional linear models. That's where machine learning (ML) comes into support. ML offers powerful methods for identifying hidden patterns and relationships within large datasets, especially when those relationships are non-linear or influenced by multiple variables [7,8]. In the context of diabetes, ML has shown strong potential in predicting disease onset, progression, and response to interventions [9–11].

Recent research further demonstrated the value of such an integrative approach. For example, one study developed an ML framework that identifies at-risk individuals using metabolic health markers, enabling more focused prevention strategies [12]. Another study introduced an explainable ML model utilizing gene expression data to identify genetic markers linked to T2DM, showcasing the potential of genomic data in shaping personalized care. A third effort combined dietary behavior with clinical biochemical data to predict T2DM onset, highlighting the critical role of lifestyle modification in prevention [13].

In addition to prediction, ML has shown promise in real-time glycemic management. For instance, Reinforcement Learning techniques have been used to optimize HbA1c prediction with accuracies up to 91 %, using continuous glucose monitoring (CGM) data and clinical variables to tailor dynamic control strategies [14]. Similarly, Random Forest algorithms have achieved 85 % accuracy in predicting poor glycemic control among medication-nonadherent individuals, emphasizing the relevance of behavioral and sociodemographic factors [15]. Feature importance analysis from these models identified key predictors such as medication adherence, variability in meal timing, and socioeconomic status.

Several studies have also focused on ML applications in diabetes care. These include models for prediabetes screening using classification algorithms, predicting peripheral arterial disease through image analysis, and estimating insulin levels using lifestyle indicators [11,16–18]. Comparative analyses have explored various ML techniques, such as neural networks, decision trees, logistic regression, and support vector machines for their performance in T2DM prediction, with logistic regression and SVM often emerging as top performers in terms of accuracy and reliability [19,20].

Although low-GI and low-GL diets are widely recognized for their role in improving glycemic control, their integration into predictive modeling remains limited. This gap presents a valuable opportunity: by incorporating these dietary metrics into machine learning (ML) frameworks, we can significantly enhance our ability to personalize diabetes prevention strategies. When combined with clinical and physiological data, individual dietary behaviors provide meaningful insights for more effective diabetes management. ML models are particularly well-suited for analyzing such complex datasets, enabling the identification of key risk factors, the prediction of glycemic responses, and the recommendation of nutrition plans tailored to each individual's unique needs [10,11,21].

This study presents an inclusive, data-driven ML framework that incorporates glycemic control indicators alongside anthropometric, nutrient intake, behavioral, and sociodemographic variables. In doing so, it aims to close the gap between dietary science and predictive modeling, offering a more holistic, interpretable, and personalized approach to diabetes prevention. By improving the accuracy and inclusivity of risk prediction, this research has the potential to inform early, targeted interventions and contribute to reducing the global burden of T2DM.

2. Methods

This study employed a comprehensive methodological framework integrating statistical analysis with advanced machine learning techniques to investigate determinants of Type 2 Diabetes Mellitus (T2DM). The approach integrated systematic data collection, preprocessing, and automated machine learning for feature engineering and predictive modeling across nutrition, anthropometry, lifestyle, and sociodemographic domains. The overall conceptual framework is illustrated in Fig. 1.

2.1. Study design

This study aimed to investigate the determinants of glycemic control among individuals diagnosed with Type 2 Diabetes Mellitus (T2DM). A cross-sectional design was employed to analyze data collected in 2022 from a sample of 3145 participants, both diabetic and non-diabetic, aged between 18 and 60 years, residing in Palestine. Participants needed to have complete information on their diet, body measurements, lifestyle, and background details. People were not included if they had type 1 or gestational diabetes, were pregnant or breastfeeding, had serious chronic or acute illnesses that could affect blood sugar control, faced difficulties in understanding or answering questions, had recently made major changes to their diet or physical activity, or were missing key information needed for the study.

A weighted sampling strategy was utilized to ensure proportional representation by gender and age, thereby enhancing the generalizability of the findings. Ethical approval was obtained from the Institutional Review Board of Al-Quds University (approval code: 129/REC/2022). Informed consent was obtained from all participants prior to their inclusion in the study.

2.2. Data collection

Data were collected from local healthcare facilities and encompassed comprehensive assessments of dietary intake, anthropometric biomarkers, lifestyle behaviors, and sociodemographic characteristics. Dietary intakes were recorded using 24-h dietary recalls, providing detailed insights into daily food consumption, including energy intake, macronutrients, micronutrients, and glycemic measures. The glycemic index and glycemic load were calculated for each food item and dish consumed, including complex Mediterranean-style cooked recipes.

Clinical measurements included documentation of diagnosed conditions and current health status, along with anthropometric data such as height, weight, and waist circumference. These measurements were used to calculate Body Mass Index (BMI) and Waist-to-Hip Ratio (WHR). Lifestyle factors were assessed through data on smoking status, eating habits, and physical activity levels, evaluated using the International Physical Activity Questionnaire (IPAQ) [22]. Sociodemographic information, including age group, gender, education, profession, family income, and region, was also collected to contextualize the findings.

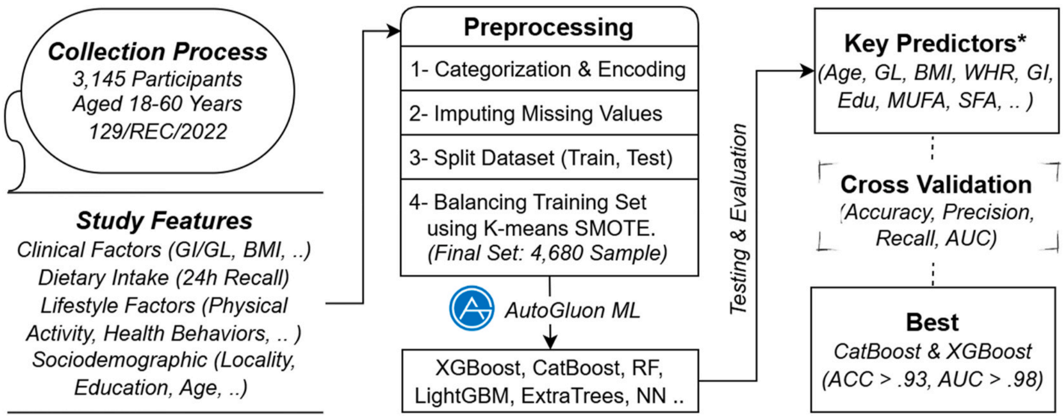

Fig. 1. Personalized T2DM prevention framework using machine learning.

2.3. Data preprocessing

The preprocessing phase involved two primary components: data normalization and missing value imputation. Continuous variables were normalized, and categorical variables were encoded to ensure compatibility with machine learning (ML) algorithms. A structured three-step imputation strategy was then applied to handle missing values. First, Cramér's V correlation analysis was used to guide grouped mode imputation, thereby preserving relationships among categorical features [23]. Second, k-nearest neighbors (KNN) imputation was applied to address remaining missing values, with careful handling of categorical-to-numerical conversions to maintain data integrity [24]. Finally, any residual missing entries were filled using column-specific mode values.

The dataset exhibited class imbalance between diabetic and non-diabetic participants, with diabetics representing the minority class. This imbalance posed a risk of biased model performance toward the majority class, potentially compromising sensitivity for diabetic case detection. Given that missing diabetic cases (false negatives) carries significant clinical consequences, addressing this imbalance was crucial for developing a reliable predictive model. Therefore, we employed a hybrid resampling strategy that combined the Synthetic Minority Oversampling Technique (SMOTE) with k-means clustering and random under-sampling, applied specifically to the training set [30]. Unlike traditional SMOTE, this enhanced approach utilized k-means clustering to identify natural groupings within the data, enabling the generation of synthetic samples that more accurately reflected the underlying distribution and reduced the risk of overgeneralization. Following preprocessing, the final dataset comprised 4680 entries.

2.4. Study features

Our study investigated potential predictor variables across four main domains: nutrition, anthropometry, lifestyle indicators, and sociodemographic agents. The status of T2DM (diabetic or healthy) served as the target variable. Table 1 presents a detailed categorization of the features object of our investigation. This comprehensive feature set enabled us to conduct a systematic analysis of the relationships between dietary patterns, sociodemographic factors, and T2DM outcomes. The breadth of these variables allowed us to develop broad population-level insights through a detailed examination of specific nutrition-diabetes relationships, facilitating the identification of key risk factors in disease development.

2.5. Nutrient metrics

As part of our broader approach to understanding the factors influencing glycemic control, we included a detailed analysis of participants' nutrient intake. This involved evaluating both macronutrients, such as carbohydrates, proteins, and fats, and key micronutrients like vitamins and minerals. These dietary components were assessed to capture the overall nutritional profile of each participant and explore their potential impact on blood sugar regulation.

Table 1
Categorization of study features: Nutrition, clinical, lifestyle, and sociodemographic domains for type 2 diabetes analysis.

Variable Domain	Category	Variables
Target Variable	*Disease Status*	*Type 2 Diabetes*
Nutrition	Macronutrients	Energy intake, Protein, Fat (SFA, MUFA, PUFA), Fiber, Cholesterol
	Micronutrients - Vitamins	Vitamin A, Carotene, Thiamin, Riboflavin, Niacin, Vitamin B6, Folate, Vitamin B12, Vitamin C
	Micronutrients - Minerals	Calcium, Phosphorus, Magnesium, Iron, Zinc, Copper, Sodium, Potassium
	Dietary Patterns	Vegetarian/Vegan status, Regular eating habits, Current diet status, Nutrition supplements
	Glycemic Measures	GI, GL
Clinical	Anthropometric Measures	BMI, WHR
	Cardiovascular Health	High cholesterol, High fat, Stroke history, Heart attack history, Bypass surgery history
	Health Status	General health status, Perceived health status
Lifestyle	Physical Activity	Physical activity levels
	Health Behaviors	Current smoking status, Regular eating habits
Sociodemographic	Geographic	Region, Locality
	Personal	Age group, Education level
	Economic	Family income, Profession

To further enrich our analysis, we also calculated two key dietary indicators: Glycemic Index (GI) and Glycemic Load (GL). These metrics allowed us to estimate how quickly and how significantly the carbohydrates in participants' diets were likely to raise blood glucose levels. By including GI and GL, we were able to assess not just how much carbohydrate someone was eating, but also the quality and metabolic effect of those carbohydrates—an important factor in managing or predicting Type 2 Diabetes. GI values for individual foods and recipes were sourced from the Eastern Mediterranean Region Food Information Data Bank (EMFID.org) [25]. Which provided standardized, regionally relevant data. These values were then used to calculate both GI and GL for each participant and incorporated directly into our machine learning models. This integration of dietary quality into our predictive framework aimed to improve the personalization and accuracy of diabetes risk profiling.

The GI was calculated using the following formula:

$$GI = \left(\frac{\textit{Area under the blood glucose curve for test food}}{\textit{Area under the blood glucose curve for reference food}}\right) \times 100 \qquad (1)$$

For GL, we used the following formula:

$$GL = \left(\frac{\textit{Carbohydrate content } (g) \textit{ of food} \times GI}{100}\right) \qquad (2)$$

2.6. Statistical and machine learning analysis

To explore the factors associated with Type 2 Diabetes Mellitus (T2DM), we first performed descriptive statistics to outline the demographic and clinical profiles of participants. We then used univariate analysis of variance (ANOVA) to compare continuous variables, such as age, BMI, and nutrient intake, between diabetic and non-diabetic groups, helping to identify potential risk indicators.

For categorical variables, we applied multinomial logistic regression, which allowed us to estimate the odds of having diabetes based on different lifestyle, behavioral, and demographic factors. This helped quantify associations and identify statistically significant predictors of diabetes status.

To build a predictive model, we used AutoGluon-Tabular [26], an automated machine learning framework that simplifies model training and evaluation. We tested nine algorithms: Random Forest, Extra Trees, LightGBM, CatBoost, XGBoost, K-Nearest Neighbors, and two types of Neural Networks (FastAI and Torch), along with a Weighted Ensemble model. We focused on ensemble methods and neural networks to capture complex, non-linear relationships between our comprehensive feature set and diabetes risk, since traditional linear models may not adequately represent the intricate interactions between nutritional, clinical, lifestyle, and sociodemographic variables. Each was fine-tuned using grid search, which optimizes model performance by testing different parameter combinations.

Model performance was validated using bagging and k-fold cross-validation [27]. Bagging improves prediction stability by averaging results from multiple models, while cross-validation ensures that every data point contributes to both training and testing, minimizing overfitting and improving generalizability.

We also implemented a stacked ensemble model, a layered approach where outputs from earlier models feed into subsequent ones. The final predictions were generated using a Greedy Weighted Ensemble, which combines multiple model predictions into a single, refined output [28].

All features listed in Table 1 (n = 48 features across nutrition, clinical, lifestyle, and sociodemographic domains) were included as model inputs without prior feature selection. This comprehensive approach was chosen to capture the full complexity of diabetes risk factors and allow the AutoGluon framework to handle feature interactions internally. To understand which features influenced the predictions most among this comprehensive set, we used perturbation-based feature importance [29]. This technique introduces noise to individual features and measures the drop in accuracy, helping us identify which variables were most critical to model performance.

This integrated approach combined traditional statistical analysis with advanced machine learning techniques, offering a reliable framework for identifying key risk factors and improving personalized prediction of T2DM.

3. Results

3.1. Statistical analysis

Table 2 shows the sociodemographic distribution of the diabetic and nondiabetic participants who partook in this research. Male participants comprised 14.3 % of the total diabetic study population, compared to 23.2 % female participants. The odds ratio (OR) for diabetic females was 1.8 (95 % CI: 1.5–2.17, **p** < 0.001). This indicated that the females were 1.8 times more likely to be diabetic compared to male participants.

Table 2
Sociodemographic of diabetic and non-diabetic participants including Proportions, distribution, and odds ratios.

Features	Diabetic n (%)	Non-Diabetic n(%)	Total Group n(%)	Odd Ratio 95 % CI
Gender				
Male	220(14.3)	1316(85.7)	1536(48.8)	1.8(1.5, 2.17) **
Female	374(23.2)	1235(76.8)	1609(51.2)	
Area of Residency				
Urban	270(45.5)	929(36.4)	1199(38.1)	0.847
Rural	196(33)	1031(40.4)	1227(39)	(0.713,1.006)*
Refugee Camp	128(21.5)	591(23.2)	719(22.9)	
Current smoking				
No	470(79.1)	1793(70.3)	2263(72)	0.684
Yes	124(20.9)	758(29.7)	882(28)	(0.507,0.922)**
Family Income				
Above Average > 600 JD	61(12.6)	290(13.1)	351(13)	0.838 (0.692,1.015)*
Average 300–600 JD	155(31.9)	826(37.4)	981(36.4)	
Below Average (<300JD)	270(55.6)	1091(49.4)	1361(50.5)	
Perceived Health Status				
Very Good	162(27.3)	1416(55.5)	1578(50.2)	1.603
Good	265(44.6)	963(37.7)	1228(39)	(1.315,1.954)**
Not Good	167(28.1)	172(6.7)	339(10.8)	
Physical activity				
No	559(94.1)	2334(91.5)	2893(92)	0.876
Yes	35(5.9)	217(8.5)	252(8)	(0.478,1.606)*
Region				
West Bank	398(67)	1604(62.9)	2002(63.7)	1.141
Gaza	196(33)	947(37.1)	1143(36.3)	(0.863,1.507)*
Educational Level				
≥12 Y	127(21.4)	1041(40.8)	1168(37.1)	1.26
9 - 11 Y	92(15.5)	642(25.2)	734(23.3)	(1.077,1.474)**
0 - 8 Y	375(63.1)	868(34)	1243(39.5)	
Age Group				
18–29	10(1.7)	1020(40)	1030(32.8)	6.288
30–39	69(11.6)	1007(39.5)	1076(34.2)	(5.296,7.467)**
40–49	215(36.2)	447(17.5)	662(21)	
50–59	300(50.5)	77(3)	377(12)	

Note: * P-value< 0.05; ** P-Value<0.001.

Participants who resided in urban areas showed higher diabetes prevalence (45.5 %) compared to residents of rural areas and refugee camps (33 % and 21.5 %, respectively). Rural residency was associated with a reduced likelihood of diabetes, as indicated by its OR of 0.847 (95 % CI: 0.713–1.006, p < 0.05). Residence in refugee camps did not deviate significantly in terms of diabetes prevalence.

Smoking proved a significant risk factor. Non-smokers showed a lower overall likelihood of having diabetes compared to their smoker counterparts (OR: 0.684, 95 % CI: 0.507–0.922, **p** < 0.001). On the other hand, participants with below-average incomes (<300 JD) showed a higher prevalence of diabetes (55.6 %) compared to those with above-average incomes (>600 JD). However, socioeconomic status merely approached, albeit did not reach, significance (OR: 0.838, 95 % CI: 0.692–1.015, p < 0.05).

"Perceived" health status was a strong predictor of diabetes (OR: 1.603, 95 % CI: 1.315–1.954, **p** < 0.001); participants who reported "Very Good" health were less likely to have diabetes compared to those who reported "Not Good" health. While physical inactivity was slightly more prevalent among diabetic participants, the association was not statistically significant (OR: 0.876, 95 % CI: 0.478–1.606, p > 0.05). Similarly, diabetes was slightly more prevalent in the West Bank compared to Gaza, although this regional difference was not statistically significant (OR: 1.141, 95 % CI: 0.863–1.507, p > 0.05).

Education level demonstrated a significant inverse relationship with the risk of diabetes (OR: 1.26, 95 % CI: 1.077–1.474, **p** < 0.001).

Participants who had fewer than 8 years of education were significantly more likely to have diabetes compared to those with 12 or more years of education. Age showed the strongest association with diabetes. Participants aged 50–59 years had the highest likelihood of diabetes, with an OR of 6.288 (95 % CI: 5.296–7.467, **p** < 0.001), compared to those aged 18–29 years.

Table 3 shows a comparison of the nutritional intake between diabetic and non-diabetic participants. The analysis here shows that the mean Glycemic Index (GI) for diabetic participants is significantly lower (87.7 ± 36.1) than that of non-diabetic participants (98.8 ± 35.5), with a total mean GI of 96.1 ± 35.9 across the study population. The observed difference between the two groups is statistically significant, as indicated by an F-statistic of 46.3 and a p-value of 0.001. This suggests that diabetic participants tend to consume foods with lower overall GI values, likely as part of dietary management strategies aimed at controlling blood glucose levels. Similarly, the Glycemic Load (GL) of diabetic participants (241 ± 180.5) is lower than that of non-diabetic participants (303.3 ± 202.7), with a total mean GL of 291.5 ± 200. This difference is also statistically significant, with an F-statistic of 48 and a p-value of 0.001. These findings indicate that non-diabetic individuals generally consume diets with higher glycemic loads compared to those with diabetes, which reflects the fewer dietary restrictions that bound this group.

The results in Table 3 also present the mean values and standard deviations for the nutrient profiles of diabetic and nondiabetic participants. Captured nutritional components included energy, macronutrients (protein, carbohydrates, fat), fatty acid composition, micronutrients, and minerals. Analyzing the statistical significance for each of these components enabled a better understanding of how differences in nutrient compositions correlated with diabetes. Diabetic individuals showed lower overall mean energy, protein, carbohydrate, and fat intakes, with significant p-values of 0.001 for energy and protein. Variations in fatty acid composition, such as higher saturated fatty acid (SFA) and monounsaturated fatty acid (MUFA) intakes among diabetics, were also notable.

3.2. Machine learning analysis

To evaluate the performance of ML classification models, we typically use metrics such as the Area Under the Curve (AUC), accuracy, sensitivity, precision (recall), and F1-score [26], with higher values indicating better all-in-all performance. Table 4 shows that XGBoost and RF achieve the highest AUC values, at 0.9863 and 0.9856, respectively, while KNN follows with an AUC of 0.9534. For accuracy, most models perform similarly, with scores around 0.929, except for CatBoost, which leads with an accuracy of 0.9333. Sensitivity is high across all nine models, with CatBoost (0.9652) and XGBoost (0.962) performing best, while KNN has the lowest sensitivity at 0.9. XGBoost and CatBoost outperform all other models used here in the precision and F1-score metrics, with WeightedEnsemble and NeuralNetTorch showing competing results. The KNN scored lower across all metrics, indicating weaker performance, while XGBoost and CatBoost demonstrated the best performance.

Fig. 2 compares the accuracy and prediction times (in seconds) on the test dataset. CatBoost achieved the highest accuracy (93.33 %) with a moderate prediction time (0.060 s), balancing accuracy and efficiency. RF, ExtraTrees, and LightGBMLarge followed closely at 92.92 % accuracy, with RF being the fastest (0.031 s) and LightGBMLarge the slowest (0.315 s). Weighted Ensemble and XGBoost also scored 92.92 %, though XGBoost had the longest prediction time (0.378 s). NeuralNet models (FastAI and Torch) had slightly lower accuracy (92.08 %) and higher prediction times, indicating reduced efficiency. KNN, while the least accurate (86.25 %), was the fastest (0.026 s), which makes it suitable for scenarios prioritizing speed over accuracy.

Fig. 3 highlights model performance based on log loss, where lower values signify better predictions. ExtraTrees showed excellent performance with the lowest log loss (0.165), followed by CatBoost (0.172) and Random Forest (0.173), demonstrating strong predictive reliability. NeuralNetTorch (0.183) and NeuralNetFastAI (0.204) performed moderately well, while XGBoost (0.254) showed a slight dip in prediction quality. LightGBMLarge (0.424) and WeightedEnsemble (0.490) had higher log loss, indicating weaker predictive accuracy. KNN recorded the highest log loss (0.813), making it the least reliable model in this analysis.

Fig. 4 illustrates the strongest predictors of T2DM, with age, GL, and BMI ranking highest. Other factors like WHR, GI, and health status also contribute strongly to the prediction.

4. Discussion

This study provides an in-depth analysis of sociodemographic, behavioral, and dietary predictive factors of Type 2 Diabetes Mellitus (T2DM). The study findings indicated significant sex-based differences in diabetes prevalence, with the female study population being 1.8 times more likely to have diabetes compared to males. This result is consistent with previous studies that reported higher prevalence rates of diabetes among women globally [30,31]. Possible explanations include hormonal fluctuations, particularly during menopause, which affect glucose metabolism, as well as social and cultural factors influencing healthcare access and dietary habits [32,33].

Rural residency was associated with a slightly reduced likelihood of diabetes occurrence (OR: 0.847). This could be attributed to a general

Table 3
Nutritional Intake Comparison Between Diabetic and Non-Diabetic Participants showing Mean Nutrient Levels and Statistical Significance.

Features	Diabetic Mean ± SD	Non-Diabetic Mean ± SD	Total Mean ± SD	F(P-Value)
Glycemic Index	87.7 ± 36.1	98.8 ± 35.5	96.1 ± 35.9	46.3(0.001)
Glycemic Load	241 ± 180.5	303.3 ± 202.7	291.5 ± 200	48(0.001)
Energy (kcal/day)	1829.1 ± 810.7	2064.2 ± 839.5	2019.8 ± 839	68.75(0.001)
Protein (g/day)	73 ± 40	77.8 ± 43.1	76.9 ± 42.6	14.27(0.001)
Carbohydrates (g/day)	235.3 ± 97.5	272.4 ± 100.1	265.4 ± 100.6	4.192(0.041)
Fat (g/day)	68.3 ± 44.9	75.9 ± 46.7	74.5 ± 46.4	5.815(0.016)
SFA (g/day)	9.3 ± 5.5	8.7 ± 4.5	8.8 ± 4.7	17.669(0.001)
MUFA (g/day)	13.1 ± 7.5	12.2 ± 5.7	12.4 ± 6.1	5.644(0.018)
Cholesterol (mg/day)	286.2 ± 281.4	287.7 ± 334.9	287.4 ± 325.4	0.204(0.652)
Vitamin B6 (mg/day)	1.3 ± 0.8	1.4 ± 0.9	1.4 ± 0.8	0.004(0.95)
Vitamin A (μg RE/day)	5672 ± 10568.9	5882.3 ± 10929	5842.6 ± 10860.6	4.58(0.032)
Vitamin B12 (μg/day)	3.1 ± 5.5	4.5 ± 14.5	4.2 ± 13.3	0(0.991)
Calcium (mg/day)	456.9 ± 323.2	486 ± 360.6	480.5 ± 354	0.604(0.437)
Phosphorus (mg/day)	952.1 ± 485.9	1027.6 ± 521.8	1013.3 ± 516	7.021(0.008)
Magnesium (mg/day)	265.4 ± 137	279.7 ± 138.5	277 ± 138.3	0.778(0.378)
Iron (mg/day)	10.8 ± 6.6	11.7 ± 6.9	11.5 ± 6.9	0.081(0.776)
Zinc (mg/day)	9.6 ± 6.5	10.1 ± 6.8	10 ± 6.8	0.143(0.705)

Table 4
Model performance metrics for glycemic control prediction in type 2 diabetes.

Family	AUC	Accuracy	Sensitivity	Precision	F1-Score
CatBoost	0.981671	0.933333	0.9652	0.8952	0.9289
XGBoost	0.986296	0.929167	0.962	0.892	0.926
RandomForest	0.985601	0.929167	0.96	0.89	0.923
ExtraTrees	0.984139	0.929167	0.958	0.888	0.921
LightGBMLarge	0.979339	0.929166	0.953	0.885	0.918
WeightedEnsemble	0.982609	0.929166	0.955	0.887	0.92
NeuralNetFastAI	0.983443	0.920833	0.95	0.88	0.914
NeuralNetTorch	0.986157	0.920833	0.951	0.882	0.916
KNeighbors	0.953391	0.8625	0.9	0.83	0.863

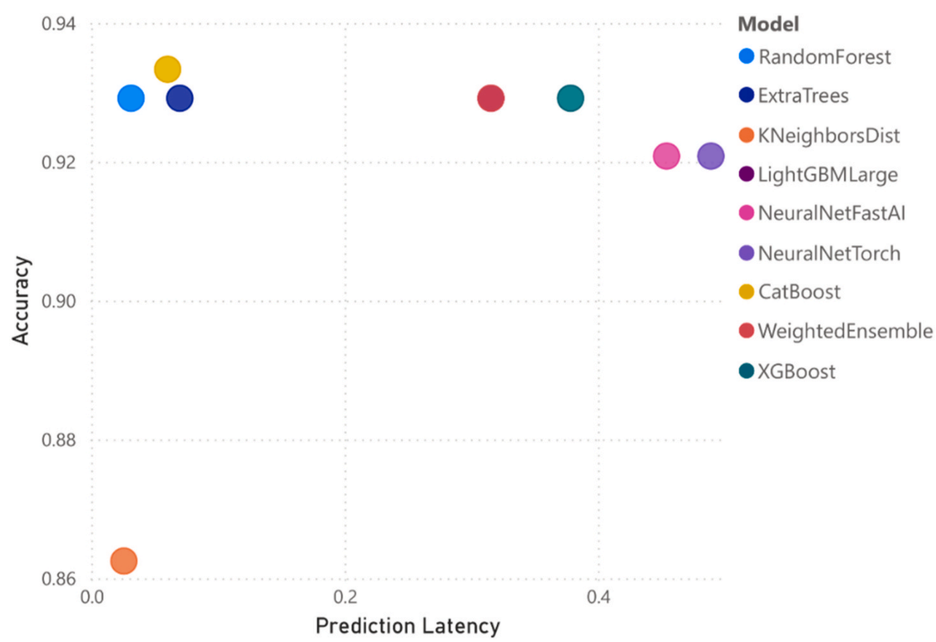

Fig. 2. Accuracy and prediction time of various models on the test dataset.

better healthcare infrastructure and greater awareness of diabetes management strategies in urban settings in the study region. Smoking emerged as a significant risk factor, with non-smokers having 32 % lower chances of diabetes compared to smokers. This aligns with findings by other studies that showed that smoking induces oxidative stress, chronic inflammation, and insulin resistance, contributing to T2DM [34,35].

A lower socioeconomic status (SES) was generally associated with higher diabetes prevalence, though it was not statistically significant. This finding is supported by Ref. [36], which indicated that lower SES limits access to nutritious food and healthcare. Education level demonstrated a strong inverse relationship with diabetes risk (OR: 1.26 for fewer than 8 years of education). This result is consistent with [37], which identified education as a key determinant of health literacy, enabling individuals to make informed dietary and lifestyle choices. Moreover, Age was the strongest predictor of diabetes, with participants aged 50–59 years having the highest T2DM occurrence odds (OR: 6.288) compared to younger participants (18–29 years). This finding is well-documented in the literature, with studies attributing it to age-related metabolic changes, reduced insulin sensitivity, and cumulative exposure to risk factors over time [30,32,33].

The dietary analysis indicated that diabetic participants had diets with significantly lower Glycemic Index (GI) and Glycemic Load (GL) values compared to non-diabetics. This finding aligns with Barclay et al. (2008), who demonstrated that low-GI diets improve glycemic control by minimizing postprandial glucose spikes [5]. The reduced GI and GL among diabetic participants likely reflect dietary modifications aimed at managing blood glucose levels. The lower intakes of energy, protein, carbohydrates, and fats among diabetic individuals further support this dietary adaptation. Studies such as [38] have shown that caloric restriction and balanced macronutrient intake are effective strategies for diabetes management. Interestingly, diabetic participants exhibited higher intakes of saturated fatty acids (SFA) and monounsaturated fatty acids (MUFA), consistent with Shah et al. (2014), who suggested that MUFAs may improve lipid profiles and glycemic control, although excessive SFA consumption warrants caution due to its potential adverse effects on cardiovascular health [39].

The ML-based analysis evaluated nine models for diabetes prediction, with CatBoost and XGBoost achieving the highest performance in terms of AUC, accuracy, sensitivity, and F1-score. CatBoost demonstrated the best overall accuracy (93.33 %) while maintaining moderate prediction times, balancing precision and efficiency. These results align with findings from Kavakiotis et al. (2017), who reported that gradient boosting algorithms excel in handling complex datasets with high-dimensional features, making them well-suited for healthcare applications [8]. Random Forest (RF) and ExtraTrees also performed strongly, with RF being the fastest model. These results are supported by Jose et al. (2024), who highlighted the efficiency and robustness of RF in predictive analytics [11]. On the other hand, KNN exhibited the lowest performance across all metrics, indicating its limited applicability in complex healthcare datasets.

The feature importance analysis presented in Fig. 3 highlights the most influential predictors in the machine learning model for Type 2 Diabetes Mellitus (T2DM). Age emerged as the most significant factor,

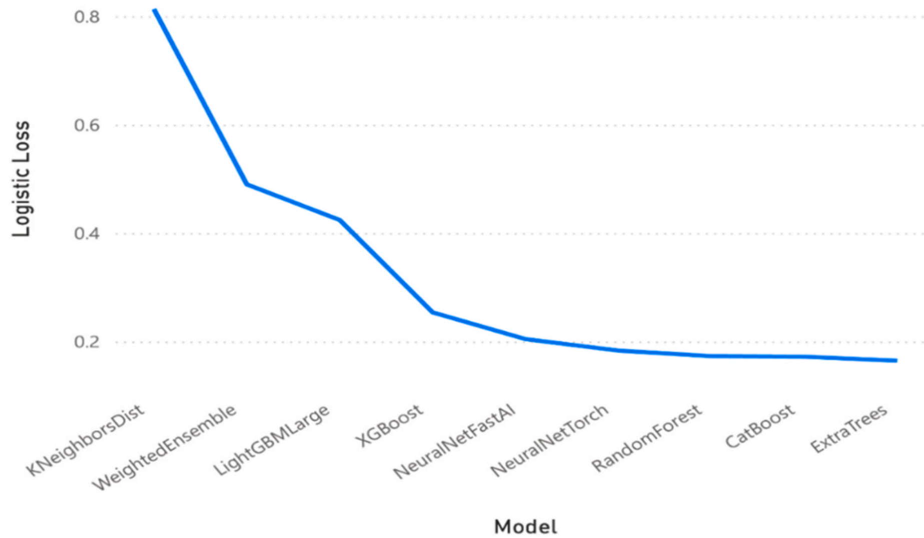

Fig. 3. Performance based on log loss.

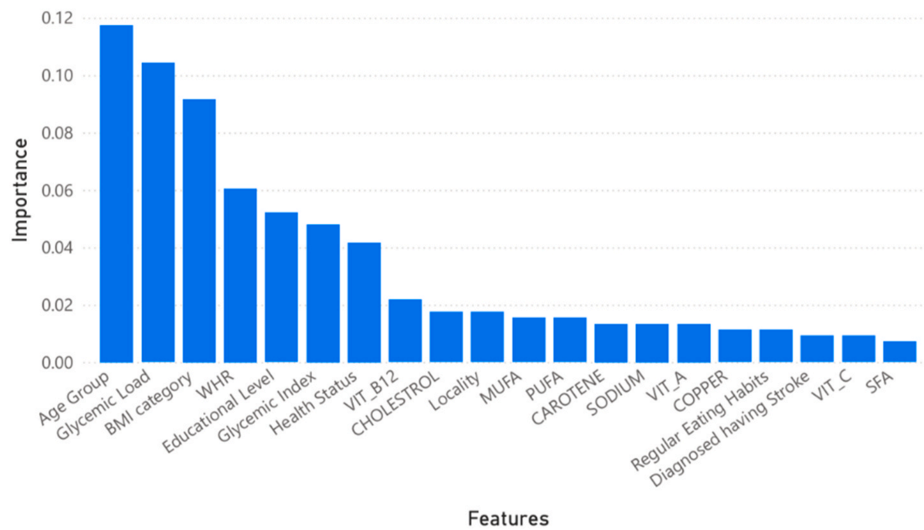

Fig. 4. Feature importance analysis.

consistent with well-established epidemiological findings that the risk of T2DM increases with advancing age due to declining insulin sensitivity and pancreatic β-cell function. Glycemic Load (GL**)** ranked second, underscoring the critical role of dietary carbohydrate quality and quantity in influencing postprandial glucose responses and long-term glycemic regulation. Body Mass Index (BMI) was also among the top predictors, reaffirming the strong link between obesity and insulin resistance, a primary driver of T2DM.

Beyond these leading variables, other factors like Waist-to-Hip Ratio (WHR) and Glycemic Index (GI) also demonstrated substantial predictive power. WHR is a known marker of central obesity and metabolic risk, often correlating more closely with T2DM risk than BMI alone. GI, while slightly less prominent than GL, reflects the rapidity of carbohydrate absorption and still plays an important role in glycemic control. Lastly, self-reported health status contributed meaningfully to the model, suggesting that perceived health may align with actual metabolic risk and could serve as a valuable, low-cost screening tool in broader populations. These findings are consistent with prior studies that emphasize the progressive decline in insulin sensitivity with aging. GL and BMI were also prominent predictors, reflecting the relationship between dietary and anthropometric factors in diabetes risk. These findings align with Malik et al. (2013), who demonstrated the critical role of dietary habits and obesity in diabetes development [12,40].

5. Conclusions

This study showed how machine learning can help us better understand and predict the risk of developing Type 2 Diabetes Mellitus (T2DM) by looking at a wide range of personal, lifestyle, and dietary factors. By combining common health indicators, like age, body mass index (BMI), and waist-to-hip ratio, with detailed information about the types of food people eat, we were able to build a model that offers a more personalized and accurate way to assess diabetes risk.

One of the key strengths of this study was the inclusion of dietary patterns based on glycemic index (GI) and glycemic load (GL), two measures that reflect how different foods and recipes affect blood sugar levels. The results showed that GL, in particular, was one of the strongest predictors of diabetes risk, alongside age and BMI. This highlights the importance of not just how much carbohydrate people consume, but also the quality and source of those carbohydrates. Foods and meals with a high glycemic load can raise blood sugar more sharply, making them especially relevant in diabetes prevention strategies. Other factors such as GI, waist-to-hip ratio, and even how individuals perceive their own health also contributed meaningfully to the predictions. Together, these findings emphasize that effective prevention should take into account both clinical and everyday lifestyle data, including what people eat, how their body stores fat, and how they view their health. This research supports the growing efforts toward personalized, data-driven health care. It shows that machine learning can help us go beyond general guidelines and move toward more tailored approaches, where interventions are informed by a person's unique combination of diet, body, and behavior.

6. Strengths and limitations

This study shows the potential and significance of integrating dietary metrics into machine learning models to support personalized glycemic control in individuals with Type 2 Diabetes Mellitus (T2DM). By addressing lifestyle-based management strategies, this work fills a gap in non-pharmacological diabetes care. Generic recommendations for lifestyle and dietary changes are often not tailored to individuals, especially when socioeconomic and educational factors influence their behaviors and food choices. However, several limitations should be noted. First, the cross-sectional design does not establish causality between dietary factors and diabetes risk. Diabetic participants may have already modified their dietary behaviors following diagnosis, adopting diets with a lower glycemic index and glycemic load as part of their diabetes management strategies. Second, the reliance on self-reported dietary data introduces potential recall bias, which may affect the accuracy of nutritional assessments. Lastly, the geographic specificity of the dataset (the West Bank and Gaza) may limit the generalizability of the findings. The dietary habits and cultural factors specific to this region may not reflect those in other populations. Future research should validate these models in more diverse populations, conduct longitudinal studies to establish temporal relationships between dietary factors and diabetes onset, and investigate their ability to predict long-term outcomes such as complications and mortality.

CRediT authorship contribution statement

Radwan Qasrawi: Writing – original draft, Methodology, Conceptualization. **Suliman Thwib:** Writing – original draft, Software, Methodology, Formal analysis. **Ghada Issa:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Razan Abu Ghoush:** Writing – review & editing, Investigation, Formal analysis. **Malak Amro:** Writing – review & editing, Investigation, Data curation, Conceptualization.

Ethical Considerations

This study complied with the Declaration of Helsinki and was approved by the Al-Quds University Institutional Review Board (Approval Code: 129/REC/2022). Informed consent was obtained from all participants, ensuring voluntary participation and the right to withdraw without penalty. Data were anonymized, securely stored, and handled in compliance with regional and international privacy regulations.

Data availability

Data will be available under reasonable requests.

AI tools declaration

During the preparation of this work, the authors used Chat GPT 4.0 to improve readability and language. The authors take full responsibility for the content of the publication.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We extend our sincere gratitude to all participating healthcare professionals, research assistants, and study participants across Palestine whose contributions made this research possible.

References

[1] M.A.B. Khan, M.J. Hashim, J.K. King, R.D. Govender, H. Mustafa, J Al Kaabi, Epidemiology of Type 2 diabetes - global burden of disease and forecasted trends, J Epidemiol Glob Health 10 (2020) 107–111, https://doi.org/10.2991/JEGH.K.191028.001.
[2] S. Templer, S. Abdo, T. Wong, Preventing diabetes complications, Intern. Med. J. 54 (2024) 1264–1274, https://doi.org/10.1111/imj.16455.
[3] B.P. Kovatchev, Metrics for glycaemic control-from HbA1c to continuous glucose monitoring, Nat. Rev. Endocrinol. 13 (2017) 425–436, https://doi.org/10.1038/nrendo.2017.3.
[4] G.E. Umpierrez, P. Kovatchev B. Glycemic Variability, How to measure and its clinical implication for type 2 diabetes, Am. J. Med. Sci. 356 (2018) 518–527, https://doi.org/10.1016/j.amjms.2018.09.010.
[5] A.W. Barclay, P. Petocz, J. McMillan-Price, V.M. Flood, T. Prvan, P. Mitchell, et al., Glycemic index, glycemic load, and chronic disease risk—a meta-analysis of observational studies, Am. J. Clin. Nutr. 87 (2008) 627–637.
[6] E. Gerontiti, A. Shalit, K. Stefanaki, P. Kazakou, D.S. Karagiannakis, M. Peppa, et al., The role of low glycemic index and load diets in medical nutrition therapy for type 2 diabetes: an update, Hormones (Basel) (2024), https://doi.org/10.1007/s42000-024-00566-7.
[7] B.J. Erickson, F. Kitamura, Magician's corner: 9. Performance metrics for machine learning models, Radiol Artif Intell 3 (2021) e200126, https://doi.org/10.1148/ryai.2021200126.
[8] I. Kavakiotis, O. Tsave, A. Salifoglou, N. Maglaveras, I. Vlahavas, I. Chouvarda, Machine learning and data mining methods in diabetes research, Comput. Struct. Biotechnol. J. 15 (2017) 104–116.
[9] Horestani FJ, Mehdi Owrang M. Predicting Diabetes with Machine Learning Analysis of Income and Health Factors. n.d.
[10] K.J. Rani, Diabetes prediction using machine learning, International Journal of Scientific Research in Computer Science, Engineering and Information Technology (2020) 294–305, https://doi.org/10.32628/CSEIT206463.
[11] R. Jose, F. Syed, A. Thomas, M. Toma, Cardiovascular health management in diabetic patients with machine-learning-driven predictions and interventions, Appl. Sci. 14 (2024), https://doi.org/10.3390/app14052132.
[12] S.M. Ganie, M.B. Malik, T. Arif, Performance analysis and prediction of type 2 diabetes mellitus based on lifestyle data using machine learning approaches, J. Diabetes Metab. Disord. 21 (2022) 339–352, https://doi.org/10.1007/s40200-022-00981-w.
[13] Pal S, Bajpayee A, Dewdhari T, Sheikh S, Thakur S. Integrating Deep Learning and Genetic Data to Detect and Predict Hereditary Transfer of Sickle Cell Anemia a Review. n.d.
[14] G. Wang, X. Liu, Z. Ying, G. Yang, Z. Chen, Z. Liu, et al., Optimized glycemic control of type 2 diabetes with reinforcement learning: a proof-of-concept trial, Nat. Med. 29 (2023) 2633–2642, https://doi.org/10.1038/s41591-023-02552-9.
[15] E. Riveros Perez, B. Avella-Molano, Learning from the machine: is diabetes in adults predicted by lifestyle variables? A retrospective predictive modelling study of NHANES 2007-2018, BMJ Open 15 (2025), https://doi.org/10.1136/bmjopen-2024-096595.
[16] E.A. Pustozerov, A.S. Tkachuk, E.A. Vasukova, A.D. Anopova, M.A. Kokina, I. V. Gorelova, et al., Machine learning approach for postprandial blood glucose prediction in gestational diabetes mellitus, IEEE Access 8 (2020) 219308–219321, https://doi.org/10.1109/ACCESS.2020.3042483.
[17] M. Lugner, A. Rawshani, E. Helleryd, B. Eliasson, Identifying top ten predictors of type 2 diabetes through machine learning analysis of UK Biobank data, Sci. Rep. 14 (2024), https://doi.org/10.1038/s41598-024-52023-5.
[18] Harnessing Machine Learning Models for Non-invasive Pre-diabetes Screening in Children and Adolescents. n.d.
[19] L.C. Padierna, L.F. Amador-Medina, B.O. Murillo-Ortiz, C. Villaseñor-Mora, Classification method of peripheral arterial disease in patients with type 2 diabetes mellitus by infrared thermography and machine learning, Infrared Phys. Technol. 111 (2020) 103531.

[20] Y. Fan, E. Long, L. Cai, Q. Cao, X. Wu, R. Tong, Machine learning approaches to predict risks of diabetic complications and poor glycemic control in nonadherent type 2 diabetes, Front. Pharmacol. 12 (2021), https://doi.org/10.3389/fphar.2021.665951.
[21] A. Massaro, N. Magaletti, G. Cosoli, A. Leogrande, F. Cannone, Use of Machine Learning to Predict the Glycemic Status of Patients with Diabetes, MDPI AG, 2023, p. 11, https://doi.org/10.3390/iech2022-12293.
[22] C.L. Craig, A.L. Marshall, M. Sjöström, A.E. Bauman, M.L. Booth, B.E. Ainsworth, et al., International physical activity questionnaire: 12-country reliability and validity, Med. Sci. Sports Exerc. 35 (2003) 1381–1395, https://doi.org/10.1249/01.MSS.0000078924.61453.FB.
[23] H. Akoglu, User's guide to correlation coefficients, Turk J Emerg Med 18 (2018) 91, https://doi.org/10.1016/J.TJEM.2018.08.001.
[24] L. Beretta, A. Santaniello, Nearest neighbor imputation algorithms: a critical evaluation, BMC Med. Inf. Decis. Making 16 (2016) 197–208, https://doi.org/10.1186/S12911-016-0318-Z/TABLES/5.
[25] EMFID, Eastern Mediterranean Region food information Data Bank n.d. https://emfid.org/frontend/web/index.php?r=site/goals. (Accessed 16 December 2024).
[26] N. Erickson, J. Mueller, A. Shirkov, H. Zhang, P. Larroy, M. Li, et al., AutoGluon-Tabular: Robust and Accurate Automl for Structured Data, 2020.
[27] P. Refaeilzadeh, L. Tang, H. Liu, Cross-Validation. Encyclopedia of Database Systems, Springer US, Boston, MA, 2009, pp. 532–538, https://doi.org/10.1007/978-0-387-39940-9_565.
[28] C.F. Kurz, W. Maier, C. Rink, A greedy stacking algorithm for model ensembling and domain weighting, BMC Res. Notes 13 (2020) 70, https://doi.org/10.1186/s13104-020-4931-7.
[29] J.R. Anaraki, H. Usefi, A Feature Selection based on perturbation theory, Expert Syst. Appl. 127 (2019) 1–8, https://doi.org/10.1016/j.eswa.2019.02.028.
[30] A. Kautzky-Willer, J. Harreiter, G. Pacini, Sex and gender differences in risk, pathophysiology and complications of type 2 diabetes mellitus, Endocr. Rev. 37 (2016) 278–316, https://doi.org/10.1210/er.2015-1137.
[31] J. Harreiter, A. Kautzky-Willer, Sex and gender differences in prevention of type 2 diabetes, Front. Endocrinol. 9 (2018), https://doi.org/10.3389/fendo.2018.00220.
[32] T. Muka, E. Asllanaj, N. Avazverdi, L. Jaspers, N. Stringa, J. Milic, et al., Age at natural menopause and risk of type 2 diabetes: a prospective cohort study, Diabetologia 60 (2017) 1951–1960, https://doi.org/10.1007/s00125-017-4346-8.
[33] F. Mauvais-Jarvis, J.A.E. Manson, J.C. Stevenson, V.A. Fonseca, Menopausal hormone therapy and type 2 diabetes prevention: evidence, mechanisms, and clinical implications, Endocr. Rev. 38 (2017) 173–188, https://doi.org/10.1210/er.2016-1146.
[34] Tentolouris N, Andrianakos A, Karanikolas G, Karamitsos D, Trontzas P, Krachtis P, et al. Type 2 Diabetes Mellitus is Associated with Obesity, Smoking and Low Socioeconomic Status in Large and Representative Samples of Rural, Urban, and Suburban Adult Greek Populations. vol. 2012. n.d.
[35] F. Xu, X.-M. Yin, M. Zhang, E. Leslie, R. Ware, N. Owen, Family average income and diagnosed type 2 diabetes in urban and rural residents in regional mainland China, Diabet. Med. 23 (2006) 1239–1246.
[36] E. Agardh, P. Allebeck, J. Hallqvist, T. Moradi, A. Sidorchuk, Type 2 diabetes incidence and socio-economic position: a systematic review and meta-analysis, Int. J. Epidemiol. 40 (2011) 804–818.
[37] S. Stringhini, C. Carmeli, M. Jokela, M. Avendaño, P. Muennig, F. Guida, et al., Socioeconomic status and the 25× 25 risk factors as determinants of premature mortality: a multicohort study and meta-analysis of 1· 7 million men and women, Lancet 389 (2017) 1229–1237.
[38] L. Schwingshackl, C. Schwedhelm, G. Hoffmann, A.-M. Lampousi, S. Knüppel, K. Iqbal, et al., Food groups and risk of all-cause mortality: a systematic review and meta-analysis of prospective studies, Am. J. Clin. Nutr. 105 (2017) 1462–1473.
[39] A.M. Shah, W. Yang, H. Mohamed, Y. Zhang, Y. Song, Microbes: a hidden treasure of polyunsaturated fatty acids, Front. Nutr. 9 (2022), https://doi.org/10.3389/fnut.2022.827837.
[40] S.M. Ganie, M.B. Malik, An ensemble Machine Learning approach for predicting Type-II diabetes mellitus based on lifestyle indicators, Healthcare Analytics 2 (2022), https://doi.org/10.1016/j.health.2022.100092.